

COMPUTER NETWORKS LAB REPORT

Student: Gaukhar Assyrzhanova

Group: IIoT-2401

Labs: 11.5.5 Subnet an IPv4 Network | 11.9.3 VLSM Design and Implementation

Lab 11.5.5 Subnet an IPv4 Network

Part 1: Design an IPv4 Subnetting Scheme

Base network: 192.168.0.0/24. Requirements: LAN-A needs minimum 50 hosts, LAN-B needs minimum 40 hosts, plus at least 2 additional unused subnets for future expansion.

Q: How many host addresses are needed in the largest required subnet?

A: 50 this is the LAN-A requirement.

Q: What is the minimum number of subnets required?

A: 4 LAN-A + LAN-B + 2 reserved for future expansion.

Q: What is the /24 subnet mask in binary?

A: 11111111.11111111.11111111.00000000

Q: In the network mask, what do the ones represent?

A: The network portion they identify which part of the address belongs to the network.

Q: In the network mask, what do the zeros represent?

A: The host portion they identify individual devices within that network.

Subnet Mask Analysis Table

Prefix	Decimal Mask	Subnets	Usable Hosts
/25	255.255.255.128	2	126
/26	255.255.255.192	4	62
/27	255.255.255.224	8	30
/28	255.255.255.240	16	14
/29	255.255.255.248	32	6
/30	255.255.255.252	64	2

Q: Which subnet masks meet the required minimum of 50 host addresses?

A: /25 (126 hosts) and /26 (62 hosts).

Q: Which subnet masks meet the minimum number of 4 subnets?

A: /26 (4), /27 (8), /28 (16), /29 (32), /30 (64).

Q: Which subnet mask meets BOTH requirements?

A: /26 (255.255.255.192) 4 subnets with 62 usable hosts each. The most efficient choice.

Resulting Subnet Table

Subnet Address	Prefix	Subnet Mask
192.168.0.0	/26	255.255.255.192
192.168.0.64	/26	255.255.255.192
192.168.0.128	/26	255.255.255.192
192.168.0.192	/26	255.255.255.192

Complete Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
CustomerRouter	G0/0	192.168.0.1	255.255.255.192	N/A
CustomerRouter	G0/1	192.168.0.65	255.255.255.192	N/A
CustomerRouter	S0/1/0	209.165.201.2	255.255.255.252	N/A
LAN-A Switch	VLAN 1	192.168.0.2	255.255.255.192	192.168.0.1
LAN-B Switch	VLAN 1	192.168.0.66	255.255.255.192	192.168.0.65
PC-A	NIC	192.168.0.62	255.255.255.192	192.168.0.1
PC-B	NIC	192.168.0.126	255.255.255.192	192.168.0.65
ISPRouter	G0/0	209.165.200.225	255.255.255.224	N/A
ISPRouter	S0/1/0	209.165.201.1	255.255.255.252	N/A
ISPSwitch	VLAN 1	209.165.200.226	255.255.255.224	209.165.200.225
ISP Workstation	NIC	209.165.200.235	255.255.255.224	209.165.200.225
ISP Server	NIC	209.165.200.240	255.255.255.224	209.165.200.225

Part 2: Device Configuration

Step 1 CustomerRouter

```
enable
configure terminal
hostname CustomerRouter
enable secret Class123
line console 0
password Cisco123
login
exit
interface GigabitEthernet 0/0
ip address 192.168.0.1 255.255.255.192
no shutdown
exit
interface GigabitEthernet 0/1
ip address 192.168.0.65 255.255.255.192
no shutdown
exit
end
```

copy running-config startup-config



The screenshot shows a terminal window titled "CustomerRouter" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying a series of commands and system messages. The output includes status messages for interface changes, a list of configuration commands entered, and a confirmation message for saving the configuration to the startup-config file. At the bottom of the terminal, there are "Copy" and "Paste" buttons, and a "Top" link.

```
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed
state to up

Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname CustomerRouter
CustomerRouter(config)#enable secret Class123
CustomerRouter(config)#line console 0
CustomerRouter(config-line)#password Cisco123
CustomerRouter(config-line)#login
CustomerRouter(config-line)#exit
CustomerRouter(config)#interface GigabitEthernet 0/0
CustomerRouter(config-if)#ip address 192.168.0.1 255.255.255.192
CustomerRouter(config-if)#no shutdown

CustomerRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0,
changed state to up

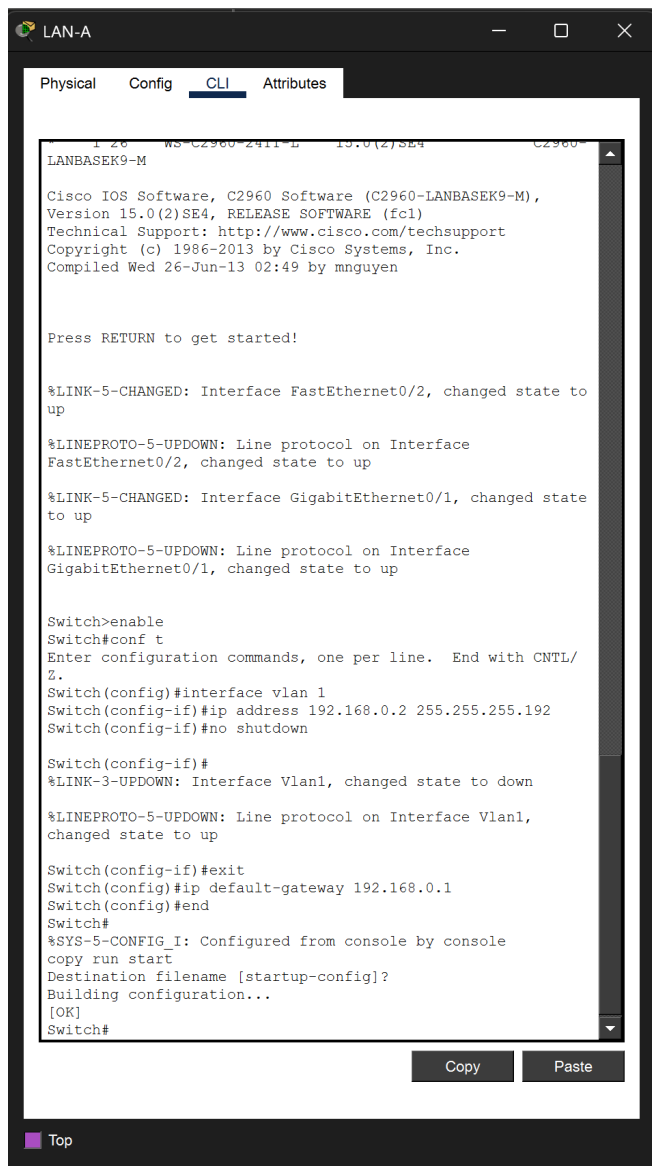
CustomerRouter(config-if)#exit
CustomerRouter(config)#interface GigabitEthernet 0/1
CustomerRouter(config-if)#ip address 192.168.0.65 255.255.255.192
CustomerRouter(config-if)#no shutdown

CustomerRouter(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

CustomerRouter(config-if)#exit
CustomerRouter(config)#end
CustomerRouter#
%SYS-5-CONFIG I: Configured from console by console
copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
CustomerRouter#
```

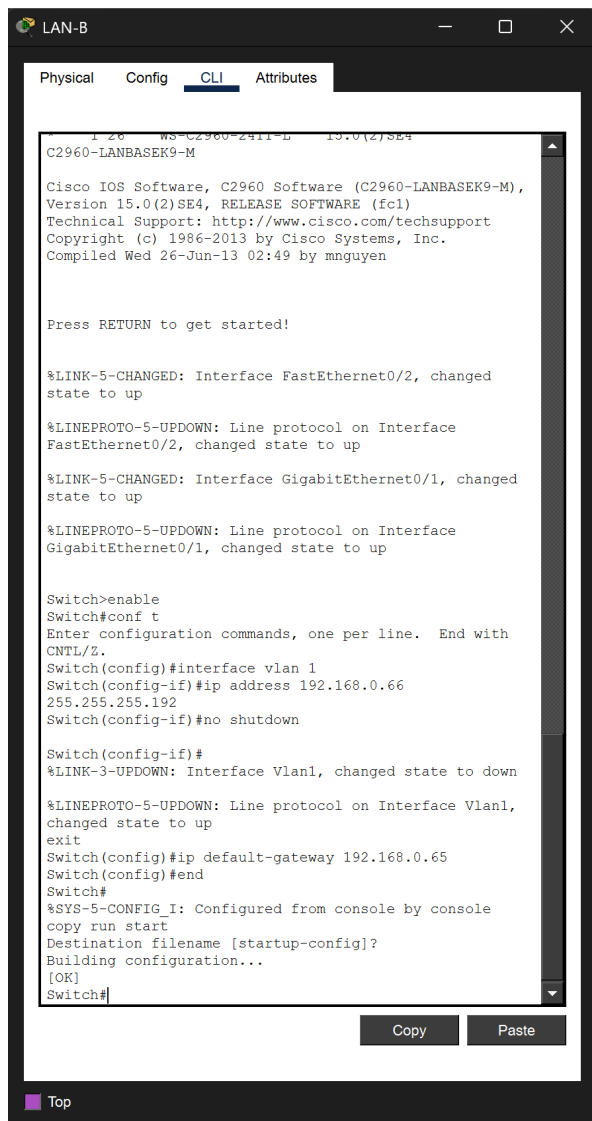
Step 2 LAN-A Switch

```
enable
configure terminal
interface vlan 1
ip address 192.168.0.2 255.255.255.192
no shutdown
exit
ip default-gateway 192.168.0.1
end
copy running-config startup-config
```



Step 3 LAN-B Switch

```
enable
configure terminal
interface vlan 1
ip address 192.168.0.66 255.255.255.192
no shutdown
exit
ip default-gateway 192.168.0.65
end
copy running-config startup-config
```



Step 4 PC-A

Field	Value
IP Address	192.168.0.62
Subnet Mask	255.255.255.192
Default Gateway	192.168.0.1

Step 5 PC-B

Field	Value
IP Address	192.168.0.126
Subnet Mask	255.255.255.192
Default Gateway	192.168.0.65

Part 3: Testing and Troubleshooting

Q: Can PC-A communicate with its default gateway?

A: Yes. Ping to 192.168.0.1 returns 4/4 replies, 0% loss.

Q: Can PC-B communicate with its default gateway?

A: Yes. Ping to 192.168.0.65 returns 4/4 replies, 0% loss.

Q: Can PC-A communicate with PC-B?

A: Yes. After initial ARP timeout, second attempt returns 4/4 replies, 0% loss. Full end-to-end connectivity confirmed.

```
PC-A
-----
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255
Reply from 192.168.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.0.126

Pinging 192.168.0.126 with 32 bytes of data:

Request timed out.
Reply from 192.168.0.126: bytes=32 time<1ms TTL=127
Reply from 192.168.0.126: bytes=32 time<1ms TTL=127
Reply from 192.168.0.126: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.0.126:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.0.126

Pinging 192.168.0.126 with 32 bytes of data:

Reply from 192.168.0.126: bytes=32 time<1ms TTL=127
Reply from 192.168.0.126: bytes=32 time<1ms TTL=127
Reply from 192.168.0.126: bytes=32 time<1ms TTL=127
Reply from 192.168.0.126: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.0.126:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

PC-B
-----
Physical  Config  Desktop  Programming  Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.65

Pinging 192.168.0.65 with 32 bytes of data:

Reply from 192.168.0.65: bytes=32 time<1ms TTL=255
Reply from 192.168.0.65: bytes=32 time<1ms TTL=255
Reply from 192.168.0.65: bytes=32 time<1ms TTL=255
Reply from 192.168.0.65: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.0.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.0.62

Pinging 192.168.0.62 with 32 bytes of data:

Reply from 192.168.0.62: bytes=32 time<1ms TTL=127
Reply from 192.168.0.62: bytes=32 time<1ms TTL=127
Reply from 192.168.0.62: bytes=32 time<1ms TTL=127
Reply from 192.168.0.62: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.0.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Lab 11.9.3 VLSM Design and Implementation

Part 1: Examine the Network Requirements

Base network: 192.168.72.0/24. Unlike Lab 11.5.5 where all subnets use the same mask, VLSM allows each subnet to have a different size only allocating as much address space as needed.

The screenshot shows the Cisco Packet Tracer interface. On the left, a network topology is displayed with a central router (Remote-Site1) connected to four switches (Sw1, Sw2, Sw3, Sw4). Sw1 and Sw2 are connected to Remote-Site1, Sw3 and Sw4 are connected to Remote-Site2. Sw1 is connected to Sw2, Sw2 to Sw3, and Sw3 to Sw4. The topology includes several hosts: User-1 (7 Hosts), User-2 (15 Hosts), User-3 (29 Hosts), and User-4 (58 Hosts). The base network is 192.168.72.0/24. On the right, an 'Instructions' window is open, titled 'Part 1: Examine the Network Requirements'. It contains two steps: Step 1: Determine the number of subnets needed, and Step 2: Determine the subnet mask information for each subnet. The instructions list the requirements for each switch: Sw1 LAN will require 7 host IP addresses, Sw2 LAN will require 15 host IP addresses, Sw3 LAN will require 29 host IP addresses, and Sw4 LAN will require 58 host IP addresses. It also asks for the number of subnets needed and the subnet mask information for each subnet.

Q: How many subnets are needed?

A: 5 subnets four LANs plus one WAN point-to-point link.

Subnet Mask Calculations

Formula: $2^n - 2 \geq \text{required hosts}$, where n = number of host bits.

LAN	Hosts Needed	Calculation	Mask	Hosts Available
Sw4	58	$2^6 - 2 = 62 \geq 58$ ✓	/26 (255.255.255.192)	62
Sw3	29	$2^5 - 2 = 30 \geq 29$ ✓	/27 (255.255.255.224)	30
Sw2	15	$2^4 - 2 = 14 < 15$ ✗ → /27	/27 (255.255.255.224)	30
Sw1	7	$2^4 - 2 = 14 \geq 7$ ✓	/28 (255.255.255.240)	14
WAN	2	$2^2 - 2 = 2 \geq 2$ ✓	/30 (255.255.255.252)	2

Note: Sw2 needs 15 hosts but /28 only provides 14 usable addresses, so /27 (30 hosts) must be used instead.

Part 2: VLSM Addressing Scheme

Subnet Table (Largest to Smallest)

Subnet Description	Hosts	Network/CIDR	First Usable	Broadcast
Sw4 LAN	58	192.168.72.0/26	192.168.72.1	192.168.72.63
Sw3 LAN	29	192.168.72.64/27	192.168.72.65	192.168.72.95
Sw2 LAN	15	192.168.72.96/27	192.168.72.97	192.168.72.127
Sw1 LAN	7	192.168.72.128/28	192.168.72.129	192.168.72.143
WAN Link	2	192.168.72.144/30	192.168.72.145	192.168.72.147

Complete Addressing Table

Rules: Router = first usable address, Switch = second usable, PC = last usable. WAN: Remote-Site1 = first, Remote-Site2 = last.

Device	Interface	IP Address	Subnet Mask	Default Gateway
Remote-Site1	G0/0	192.168.72.129	255.255.255.240	N/A
Remote-Site1	G0/1	192.168.72.97	255.255.255.224	N/A
Remote-Site1	S0/0/0	192.168.72.145	255.255.255.252	N/A
Remote-Site2	G0/0	192.168.72.65	255.255.255.224	N/A
Remote-Site2	G0/1	192.168.72.1	255.255.255.192	N/A
Remote-Site2	S0/0/0	192.168.72.146	255.255.255.252	N/A
Sw1	VLAN 1	192.168.72.130	255.255.255.240	192.168.72.129
Sw2	VLAN 1	192.168.72.98	255.255.255.224	192.168.72.97
Sw3	VLAN 1	192.168.72.66	255.255.255.224	192.168.72.65
Sw4	VLAN 1	192.168.72.2	255.255.255.192	192.168.72.1
User-1	NIC	192.168.72.142	255.255.255.240	192.168.72.129
User-2	NIC	192.168.72.126	255.255.255.224	192.168.72.97
User-3	NIC	192.168.72.94	255.255.255.224	192.168.72.65
User-4	NIC	192.168.72.62	255.255.255.192	192.168.72.1

Part 3: Device Configuration

Step 1 Remote-Site1 Router

```
enable
configure terminal
interface GigabitEthernet 0/0
ip address 192.168.72.129 255.255.255.240
no shutdown
exit
interface GigabitEthernet 0/1
ip address 192.168.72.97 255.255.255.224
no shutdown
exit
```



```
end
copy running-config startup-config
```



```
Remote-Site1
CLI

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 192.168.72.146 (Serial0/0/0) is up: new adjacency

Remote-Site1>enable
Remote-Site1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Remote-Site1(config)#interface GigabitEthernet 0/0
Remote-Site1(config-if)#ip address 192.168.72.129 255.255.255.240
Remote-Site1(config-if)#no shutdown

Remote-Site1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Remote-Site1(config)#interface GigabitEthernet 0/1
Remote-Site1(config-if)#
Remote-Site1(config-if)#ip address 192.168.72.97 255.255.255.224
Remote-Site1(config-if)#no shutdown

Remote-Site1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Remote-Site1(config)#end
Remote-Site1#
$SYS-5-CONFIG I: Configured from console by console
copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Remote-Site1#
```

Step 2 Sw3 Switch

```
enable
configure terminal
interface vlan 1
ip address 192.168.72.66 255.255.255.224
no shutdown
exit
ip default-gateway 192.168.72.65
end
copy running-config startup-config
```

```

Sw3
CLI

TOP Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch Ports Model          SW Version        SW Image
-----
*  1 26  WS-C2960-24TT-L    15.0(2)SE4       C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Sw3>enable
Sw3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Sw3(config)#int vlan 1
Sw3(config-if)#ip address 192.168.72.66 255.255.255.224
Sw3(config-if)#no shutdown

Sw3(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Sw3(config-if)#no shutdown
Sw3(config-if)#exit
Sw3(config)#ip default-gateway 192.168.72.65
Sw3(config)#end
Sw3#
%SYS-5-CONFIG I: Configured from console by console
copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Sw3#

```

Step 3 User-4 PC

Field	Value
IP Address	192.168.72.62
Subnet Mask	255.255.255.192
Default Gateway	192.168.72.1

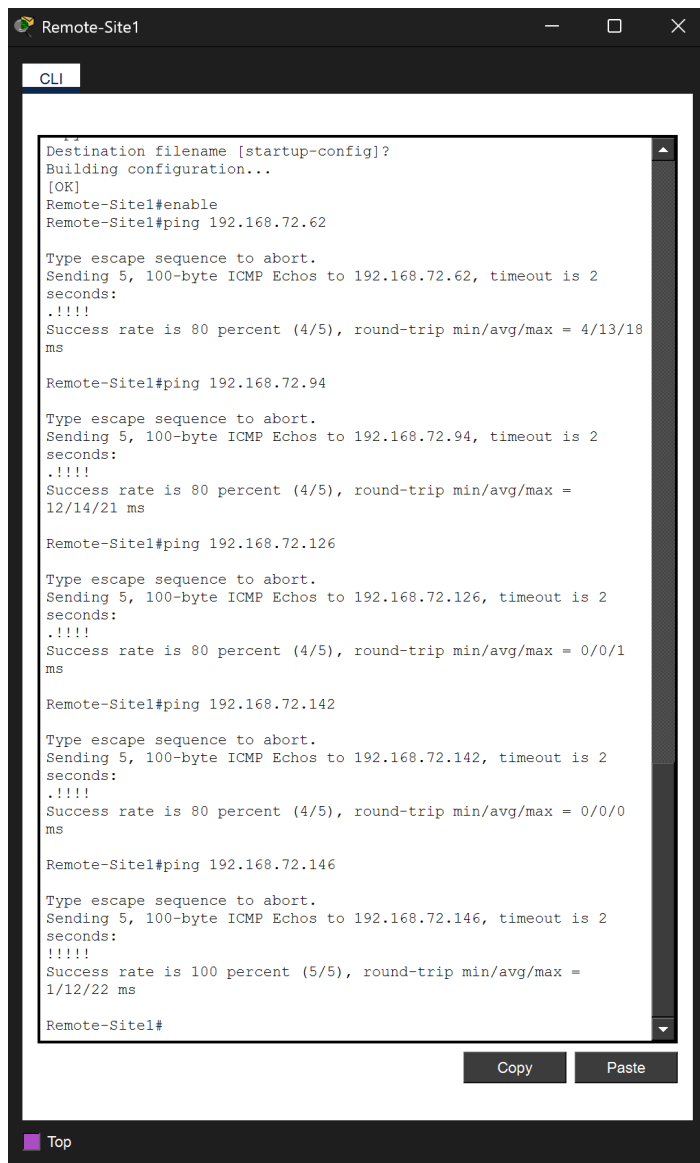
Step 4: Connectivity Verification

From Remote-Site1:

```

ping 192.168.72.62    → Success (User-4)
ping 192.168.72.94    → Success (User-3)
ping 192.168.72.126   → Success (User-2)
ping 192.168.72.142   → Success (User-1)
ping 192.168.72.146   → Success (Remote-Site2 WAN)

```



The screenshot shows a CLI window titled "Remote-Site1" with a "CLI" tab. The terminal output shows the following sequence of commands and results:

```
Destination filename [startup-config]?
Building configuration...
[OK]
Remote-Site1#enable
Remote-Site1#ping 192.168.72.62

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.72.62, timeout is 2
seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 4/13/18
ms

Remote-Site1#ping 192.168.72.94

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.72.94, timeout is 2
seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max =
12/14/21 ms

Remote-Site1#ping 192.168.72.126

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.72.126, timeout is 2
seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1
ms

Remote-Site1#ping 192.168.72.142

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.72.142, timeout is 2
seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0
ms

Remote-Site1#ping 192.168.72.146

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.72.146, timeout is 2
seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max =
1/12/22 ms

Remote-Site1#
```

At the bottom of the CLI window, there are "Copy" and "Paste" buttons, and a "Top" link.

From User-4:

```
ping 192.168.72.1      → Success (gateway)
ping 192.168.72.65     → Success (Remote-Site2 G0/0)
ping 192.168.72.97     → Success (Remote-Site1 G0/1)
ping 192.168.72.129    → Success (Remote-Site1 G0/0)
ping 192.168.72.145    → Success (Remote-Site1 WAN)
```

The image shows two side-by-side screenshots of a Windows Command Prompt window titled "User-4". The window has tabs for "Desktop" and "Programming", with "Programming" selected. The Command Prompt shows the following commands and results:

Left Window:

```
C:\>ping 192.168.72.1

Pinging 192.168.72.1 with 32 bytes of data:

Reply from 192.168.72.1: bytes=32 time<1ms TTL=255
Reply from 192.168.72.1: bytes=32 time<1ms TTL=255
Reply from 192.168.72.1: bytes=32 time<1ms TTL=255
Reply from 192.168.72.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.72.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
    loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.72.65

Pinging 192.168.72.65 with 32 bytes of data:

Reply from 192.168.72.65: bytes=32 time<1ms TTL=255
Reply from 192.168.72.65: bytes=32 time<1ms TTL=255
Reply from 192.168.72.65: bytes=32 time<1ms TTL=255
Reply from 192.168.72.65: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.72.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
    loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.72.97

Pinging 192.168.72.97 with 32 bytes of data:

Reply from 192.168.72.97: bytes=32 time=1ms TTL=254
Reply from 192.168.72.97: bytes=32 time=1ms TTL=254
Reply from 192.168.72.97: bytes=32 time=1ms TTL=254
Reply from 192.168.72.97: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.72.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
    loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

Right Window:

```
C:\>ping 192.168.72.97

Pinging 192.168.72.97 with 32 bytes of data:

Reply from 192.168.72.97: bytes=32 time=1ms TTL=254
Reply from 192.168.72.97: bytes=32 time=1ms TTL=254
Reply from 192.168.72.97: bytes=32 time=1ms TTL=254
Reply from 192.168.72.97: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.72.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
    loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 192.168.72.129

Pinging 192.168.72.129 with 32 bytes of data:

Reply from 192.168.72.129: bytes=32 time=1ms TTL=254
Reply from 192.168.72.129: bytes=32 time=1ms TTL=254
Reply from 192.168.72.129: bytes=32 time=1ms TTL=254
Reply from 192.168.72.129: bytes=32 time=24ms TTL=254

Ping statistics for 192.168.72.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
    loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 24ms, Average = 6ms

C:\>ping 192.168.72.145

Pinging 192.168.72.145 with 32 bytes of data:

Reply from 192.168.72.145: bytes=32 time=1ms TTL=254
Reply from 192.168.72.145: bytes=32 time=1ms TTL=254
Reply from 192.168.72.145: bytes=32 time=1ms TTL=254
Reply from 192.168.72.145: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.72.145:
    Packets: Sent = 4, Received = 4, Lost = 0 (0%
    loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

Conclusion

In Lab 11.5.5, we applied **fixed-length subnetting** dividing 192.168.0.0/24 into four equal /26 subnets (62 hosts each). The /26 mask was the only one that simultaneously satisfies the 50-host minimum and provides 4 subnets (2 active + 2 reserved). In Lab 11.9.3, we applied **VLSM**, which is more efficient for networks with varying LAN sizes. By ordering subnets largest-to-smallest and assigning /26, /27, /27, /28, and /30 masks, we used only 148 of 256 available addresses leaving 108 for future growth. Both labs reinforced CLI configuration skills, systematic IP address planning, and ping-based connectivity verification.